

STANDARD OPERATING PROCEDURE & COMPLETE APPLICATION USER MANUAL

Comprehensive Guide with Exact Values, Form Fields, Regulatory Calculations & Operational Workflows for All System Scenarios

Technical Authority: OIML R 76-1:2006 (E) & OIML R 76-2:2007 (E) | Legal Metrology (General) Rules, 2011

Document Version: 1.0.0 (Official)	Target Audience: Inspectors, Supervisors, RRSL Officers, Manufacturers
Effective Date: 29 September 2026	Standard Implementation: OIML R 76 Non-Automatic Weighing Instruments
System URL: http://localhost:3000 (Vite) / :8005 (FastAPI)	Database: Relational Normalized Persistence (labelguard.db)

1. System Architecture & Role-Based Access Control (RBAC)

The system strictly regulates operational actions based on user roles enforced at the FastAPI backend layer. Each user must use their official authorized credentials. Below is the operational role matrix:

Role Name	Login Email	Password	Authorized Permissions & Operations
Inspector / Test Engineer	inspector@legalmetrology.gov.in	Inspector@123	Create manufacturers, instruments, enter observations, upload photos, calculate compliance, submit draft evaluations for review.
Supervisor / Reviewer	supervisor@legalmetrology.gov.in	Supervisor@123	Technical review of observations, request corrections with mandatory notes, approve compliant reports, apply digital sign, finalize certificates.
Directorate Administrator	admin@legalmetrology.gov.in	Admin@123	Manage testing laboratories, equipment manufacturers, statutory rule versions, view immutable audit logs, system-wide overrides.
Public / Field Viewer	viewer@legalmetrology.gov.in	Viewer@123	Read-only access. Search pattern approvals, inspect approved certificates, download official PDF/DOCX reports. Blocked from write actions (HTTP 403).

2. Scenario 1: End-to-End Compliant Evaluation (Class III Bench Scale - BPM-15)

Objective: Complete testing and certification for a Class III Non-Automatic Weighing Instrument with maximum capacity 15 kg and verification scale interval $e = 5$ g. All errors remain strictly within OIML R 76 Table 6 limits.

Step 1.1: Instrument Technical Specifications & Laboratory Conditions

Field Name	Exact Value to Enter	Regulatory Clause / Metrological Rationale
Applicant Type	Manufacturer	Section 22, Legal Metrology Act, 2009
Manufacturer Name	Bharat Precision Metrology Ltd.	Registered under Legal Metrology Central Rules
Manufacturer Address	Plot 42, Okhla Industrial Area Phase-III, New Delhi	Physical manufacturing & assembly facility
Model Designation	BPM-15-DIGITAL	Model / Pattern approval identification
Instrument Type	Non-Automatic Counter Scale	OIML R 76-1 Clause 2.1 definition
Accuracy Class	Class III (Medium)	OIML R 76 Table 3 classification
Max Capacity (Max)	15.000 kg	Maximum weighing capacity without tare
Min Capacity (Min)	0.100 kg	Min = 20 e = 20 × 0.005 kg = 0.100 kg (Clause 3.4.3)
Verification Scale (e)	0.005 kg (5 g)	Scale interval used for type approval (Clause 3.2)
Actual Scale Interval (d)	0.005 kg (d = e)	Clause 3.4.1 (d = e for standard digital indicators)
Scale Intervals count (n)	3,000	Calculated: n = Max / e = 15.0 / 0.005 = 3,000 (Complies with 100 <= n <= 10,000)
Temperature Range	-10°C to +40°C	Standard temperature limit per Clause 3.9.2.1
Testing Laboratory	National Legal Metrology Type Evaluation Lab	NABL ISO/IEC 17025 Accredited & OIML Issuing Authority
Standard Weights Used	Class M1 Working Standards (NPL Traceable)	OIML R 111-1 certified reference standard masses

Step 1.2: Clause A.4.4 Weighing Performance Test Data Entry

Perform weighing with increasing loads from Zero to Max, then decreasing back to Zero. For each point, determine the changeover turning point by adding small weights $\Delta L = 0.1e$ until the indication steps up by $1d$.
Formula: $E = I + 0.5e - \Delta L - L$; Corrected error: $E_c = E - E_0$.

Dir	Load (L)	Indication (I)	Add Load (ΔL)	Corr Error (E_c)	MPE Limit	Clause / Result
INCR	0.000 kg	0.000 kg	0.0025 kg	0.0000 kg	± 0.0025 kg (0.5e)	Zero Reference / PASS
INCR	0.500 kg (100e)	0.500 kg	0.0025 kg	0.0000 kg	± 0.0025 kg (0.5e)	Table 6 Step 1 / PASS
INCR	2.500 kg (500e)	2.500 kg	0.0025 kg	0.0000 kg	± 0.0025 kg (0.5e)	500e Boundary / PASS
INCR	5.000 kg (1000e)	5.000 kg	0.0020 kg	+0.0005 kg	± 0.0050 kg (1.0e)	Table 6 Step 2 / PASS
INCR	10.000 kg (2000e)	10.000 kg	0.0015 kg	+0.0010 kg	± 0.0050 kg (1.0e)	2000e Boundary / PASS
INCR	15.000 kg (3000e)	15.000 kg	0.0010 kg	+0.0015 kg	± 0.0075 kg (1.5e)	Max Capacity / PASS
DECR	10.000 kg	10.000 kg	0.0018 kg	+0.0007 kg	± 0.0050 kg (1.0e)	Hysteresis = 0.3g / PASS
DECR	5.000 kg	5.000 kg	0.0022 kg	+0.0003 kg	± 0.0050 kg (1.0e)	Hysteresis = 0.2g / PASS
DECR	0.000 kg	0.000 kg	0.0025 kg	0.0000 kg	± 0.0025 kg (0.5e)	Zero Return / PASS

Step 1.3: Clause A.4.10 Repeatability & Clause A.4.7 Eccentricity Test Data

Test Battery	Applied Test Load	Recorded Indications / Runs	Spread / Max Difference	Permissible Limit & Status
Clause A.4.10 Repeatability	15.000 kg (Max)	Run 1: 15.000 kg Run 2: 15.005 kg Run 3: 15.000 kg Run 4: 15.000 kg Run 5: 15.005 kg	Max Diff: 0.005 kg (15.005 - 15.000)	Limit: MPE = 0.0075 kg COMPLIANT (PASS)
Clause A.4.7 Eccentricity (Corner Loading)	5.000 kg (1/3 Max)	Pos 1 (Center): 5.000 kg (E=0.0) Pos 2 (Front-Left): 5.000 kg (E=0.0) Pos 3 (Rear-Left): 5.005 kg (E=+0.5e) Pos 4 (Rear-Right): 5.000 kg (E=0.0) Pos 5 (Front-Right): 5.000 kg (E=0.0)	Max Error: +0.0025 kg (+0.5e)	Limit: MPE = 0.0050 kg (1.0e) COMPLIANT (PASS)

3. Scenario 2: Non-Compliance & Pattern Approval Rejection (Defective Load Cell)

Objective: Demonstrate how the calculation engine detects and flags excessive error violating OIML R 76 Table 6, and how the supervisor executes a statutory rejection under the Legal Metrology Act.

Parameter	Defective Test Input Value	Engine Evaluation & Non-Compliance Finding
Instrument Model	DEFECT-20-FAIL	Class III Electronic Retail Scale (Max: 20 kg, e = 5 g)
Test Load Applied	10.000 kg (2,000 e)	Boundary load between Step 2 and Step 3 MPE tier
Indicated Reading (I)	10.015 kg	Instrument displays 15 grams over nominal weight
Add-Load Turning (ΔL)	0.0025 kg (0.5 e)	Flash point changeover measurement
Calculated Error (Ec)	+0.0125 kg (+2.5 e)	E = 10.015 + 0.0025 - 0.0025 - 10.000 = +0.0150 kg; Ec = +0.0125 kg
OIML R 76 Table 6 MPE	±0.0050 kg (±1.0 e)	Maximum Permissible Error at 10.0 kg is 1.0 e (5 grams)
Engine Compliance Status	NON_COMPLIANT (FAIL)	Violation: Measured error (+12.5 g) exceeds permissible limit (5.0 g) by 250%
Supervisor Action	Click 'Reject' Button	Enter statutory rejection grounds in modal; system records immutable REJECT status

4. Scenario 3: Supervisory Technical Review & Correction Request Cycle

The workflow strictly enforces the legal hierarchy between Test Engineers and Supervisors:

Step #	Actor & Role	UI Action / Screen	State Transition & Backend Event
1	Inspector (Rajesh Sharma)	Repository Detail Drawer • Click 'Submit for Review'	Status transitions DRAFT → SUBMITTED. System logs audit record.
2	Supervisor (Priya V. Iyer)	Top Header Switcher • Switch to Supervisor Role • Click 'Begin Review'	Status transitions SUBMITTED → UNDER_REVIEW. Technical review lock engaged.
3	Supervisor (Priya V. Iyer)	Click 'Request Correction' • Enter note: 'Re-verify corner 3 eccentric loading; attach calibration certificate for F2 masses.'	Status transitions UNDER_REVIEW → CORRECTION_REQUIRED. Report returned to Inspector queue.
4	Inspector (Rajesh Sharma)	Evidence Tab • Upload calibration cert • Click 'Submit for Review'	Status transitions CORRECTION_REQUIRED → SUBMITTED. Ready for final review.
5	Supervisor (Priya V. Iyer)	Begin Review • Verify all 7 batteries PASS • Click 'Approve Report'	Status transitions UNDER_REVIEW → APPROVED_COMPLIANT.

5. Scenario 4: Digital Signing, Versioning & Document Export (PDF / Word)

Upon reaching APPROVED_COMPLIANT status, the supervisor applies a cryptographic digital signature stamp under Section 22 of the Legal Metrology Act, 2009. The document is permanently sealed into an immutable archive.

Signing Step	Required Input / Parameter	Output & Verification Metric
1. Digital Sign Modal	Signing Officer: Dr. Priya V. Iyer Designation: Assistant Controller (Metrology) Remarks: Officially certified under Sec 22	Generates 128-bit Signature Token & Cryptographic Token: VERIFIED_AUTHENTIC
2. SHA-256 Checksum	Calculated across Report #, Model, Max, e, Status, and Test Summaries	Example: 9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08
3. Finalize Certificate	Click ' Finalize & Lock Certificate ' button in Repository Drawer	Status transitions to FINALIZED . Report is locked. Creates record in report_versions.
4. PDF Export	Click ' Download PDF ' in Quick Actions Bar	Streams official printable PDF with Government Emblem, Ashoka Seal, and table batteries.
5. Word (.docx) Export	Click ' Download Word (.docx) ' in Quick Actions Bar	Streams fully editable MS Word report for departmental filing.

6. Scenario 5: Uploading Physical Evidence & Inspecting Audit Logs

Every official type approval must include photographic documentation of the instrument, its rating nameplate, and traceable calibration certificates.

Attachment Upload Rules:

- Allowed formats: PNG, JPG, JPEG, WEBP, PDF, DOCX.
- Maximum file size: 15 MB per file.
- Upload categories: INSTRUMENT_PHOTO, MARKING_PLATE, TEST_SETUP, CALIBRATION_CERTIFICATE.
- Access: Inspect through the '**Evidence**' tab in the Evaluation Drawer.

Audit Trail Verification:

Click the '**Audit Trail**' tab in the Evaluation Drawer. The system displays chronological, tamper-evident logs recording: User ID, Timestamp, Action (e.g. STATUS_TRANSITION_SUBMIT, ATTACHMENT_UPLOADED, DIGITAL_SIGNATURE_APPLIED), and previous vs. updated values.

7. Scenario 6: Packaged Commodities Rules 2011 Compliance Verification

In addition to OIML R 76 NAWI weighing instruments, the application includes the Legal Metrology (Packaged Commodities) Rules, 2011 module:

1. Click the **Module Switcher** in the top navigation bar and select '**Packaged Commodities (Rules 2011)**'.
2. Navigate to '**Verify Product**' tab.
3. Upload product packaging images (Front, Back, Side).
4. The integrated AI OCR pipeline extracts statutory declarations: **MRP (inclusive of all taxes)**, **Net Quantity (standard SI units)**, **Unit Sale Price (USP)**, **Month & Year of Manufacture**, and **Consumer Care Details**.
5. The rule engine validates each declaration deterministically against Rule 6 statutory mandates and generates a violation breakdown if prohibited units or missing declarations are detected.

OFFICIAL NOTICE: This manual has been verified against OIML Recommendation R 76-1:2006 (E), OIML R 76-2:2007 (E), and the Legal Metrology (General) Rules, 2011 (Seventh Schedule). All calculations and tolerance limits depicted herein are statutory and binding on all accredited Regional Reference Standard Laboratories (RRSL) and Type Evaluation Testing Centers under the Department of Consumer Affairs, Government of India.